

Employee Attrition Notebook — Summary & Business Conclusions

Summary of Work Done in the Notebook

- Initial import and inspection: the dataset has 1,470 rows and 35 columns, with no missing values and no duplicate rows.
- Departments and job roles: calculated employee counts and attrition rate per Department (using a crosstab with `normalize='index'`), then repeated the same approach at the JobRole level.
- Salary, experience and performance: average income per job role, correlation between experience and income, and average performance rating per department.
- Satisfaction: average of 4 satisfaction indicators, plotted as 4 subplots.
- Attrition drivers: OverTime, JobSatisfaction, and WorkLifeBalance each compared against attrition rate, plus a full correlation heatmap at the end.

Key Business Conclusions

- Overall attrition rate is 16.1% (237 of 1,470 employees left).
- Overtime is the strongest single driver of attrition: employees who work overtime leave at 30.5%, nearly 3x the rate of those who don't (10.4%). This should be the top priority for HR intervention.
- Sales is the highest-attrition department (20.6%), driven almost entirely by one role: Sales Representatives, who leave at 39.8% — by far the highest of any job role in the company, and roughly double the next-highest role (Laboratory Technician, 23.9%).
- Compensation gap likely compounds the problem: Sales Representatives also earn the lowest average monthly income of all job roles, while performance ratings across departments are nearly identical ($\sim 3.1/4$) — meaning the issue is not underperformance, but likely pay and role conditions.
- Experience and income are strongly correlated (0.77): income grows consistently with tenure, so early-career/low-tenure employees (like most Sales Reps) are structurally underpaid relative to more senior staff.
- Satisfaction has a clear inverse relationship with attrition: attrition drops from 22.8% at the lowest Job Satisfaction level to 11.3% at the highest, and from 31.2% at the worst Work-Life Balance level to 14.2% at the best.
- Recommendation: prioritize reducing overtime and improving work-life balance and compensation specifically for the Sales Representative role — this single group is the company's biggest attrition risk.